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CARIN: Wireless CSI-based Driver Activity Recognition under the Interference of Passengers

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Recent studies have proposed to use the Channel State Information (CSI) of WiFi wireless channel for human gesture recognition. As an important application, CSI-based driver activity recognition in passenger vehicles has received increasing research attention. However, a serious limitation of almost all the existing WiFi-based recognition solutions is that they can only recognize the activity of a single person at a time, because the activities of other people (if performed at the same time) can interfere with the WiFi signals. In a sharp contrast, there can often be one or more passengers in any vehicles.

In this paper, we propose CARIN, CSI-based driver Activity Recognition under the INterference of passengers. CARIN features a combination-based solution that profiles all the possible activity combinations of driver and (one or more) passengers in offline training and then performs recognition online. To attack possible combination explosion, we first leverage in-car pressure sensors to significantly reduce combinations, because there are only limited seating options in a passenger vehicle. We then formulate a distance minimization problem for fast runtime recognition. In addition, a period analysis methodology is designed based on the kNN classifier to recognize activities that have a sequence of body movements, like continuous head nodding due to driver fatigue. Our results in a real car with 3,000 real-world traces show that CARIN can achieve an overall F1 score of 90.9%, and outperforms the three state-of-the-art solutions by 32.2%.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

Additional Key Words and Phrases: Wireless Sensing, Activity Recognition, ADAS

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1 INTRODUCTION

Wireless sensing based on WiFi signals has recently shown great promise in human gesture recognition [2, 25, 27], mainly because of its 1) non-intrusive nature, 2) high recognition accuracy, and 3) use of only commercial-off-the-shelf devices. Recent studies [22, 24, 25, 31, 34] have also successfully exploited the Channel State Information (CSI) of the WiFi wireless channel, which is traditionally used by the WiFi receiver for channel monitoring, to recognize fine-grained human activities. The fundamental principle that enables CSI-based activity recognition is that the reflected WiFi signal can change when the target person moves his/her body parts. On the WiFi receiver side, those signal changes can be uniquely recognized for different human activities based on pre-trained patterns. CSI-based recognition can overcome the limitations of traditional camera-based detection methodologies, such as Line-of-Sight (LoS), brightness requirements, obscurity caused by motions [12, 20], and possible defect of image

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processing [21]. Moreover, compared with solutions using Inertial Measurement Unit (IMU) sensors [10, 13], CSI-based recognition systems are also non-intrusive and do not require the user to buy and wear a certain device on his/her body. As a result, a variety of new applications have been proposed for CSI-based recognition, such as human identification [35], breathing monitoring [26, 36], and interactive gaming [19].

Particularly, an important application area is driver activity recognition because WiFi is expected to be equipped soon in cars by replacing Bluetooth for in-car interaction with the development of Vehicle-to-Vehicle (V2V) network based on 802.11p [29]. Several recent studies have thus proposed in-car driver activity recognition for different purposes, such as entertainment control [20], fatigue detection [9], head tracking [32], and takeover safety in autonomous driving [3]. However, a serious limitation of those solutions, as well as almost all the existing WiFi-based recognition studies in general, is that they can only recognize the activity of *a single person at a time* because the activities of other people (if performed at the same time) can interfere with the WiFi signals. As a result, the WiFi receiver can fail to find a good match between the observed mixed WiFi signal changes and the pre-trained single activity patterns. This is in sharp contrast to the fact that there can often be one or more passengers in any vehicles. Therefore, the state-of-the-art driver activity recognition solutions cannot be practically implemented if this limitation is not overcome.

Existing approaches to separating mixed signals (e.g., [16, 18, 23, 32]) are mainly in three categories. First, signal separation has been well studied in radar systems. With the help of Independent Component Analysis (ICA) [8] or blind source separation [14], the changes of each signal can be extracted [16]. Unfortunately, WiFi signals are much noisier than radar signals, making it difficult for this solution to separate the changes caused by the driver and different passengers. Second, some recent studies have proposed to find the best placement of WiFi receivers, such that the passengers' interference with the driver signal can be minimized and treated as background noise [18, 32]. However, though this solution works acceptably when the passengers make just small movements (e.g., head turning), it fails when the passengers move their torsos or arms (see Section 3). Third, another recent study, called WiMU [23], has tried to find the influence of each person using frequency analysis by identifying his/her activity's duration based on sudden changes in the spectrum. However, WiMU is designed for large-scale in-room activities, such as running and walking, in which the sudden signal changes at the beginning and end of an activity can be easily captured. In a vehicle, however, most activities are much smaller-scale, which cause only much smaller changes in the wireless signal. In addition, WiMU is location-unaware because people can move freely in a room. As a result, though WiMU can recognize all the ongoing activities, it cannot tell who their respective performers are, e.g., which activity is being performed by the driver (instead of a passenger).

In this paper, we propose CARIN, CSI-based driver Activity Recognition under the **IN**terference of passengers. CARIN aims to recognize what activities the driver and passengers are performing, respectively, even when they perform those activities together at the same time. The major challenge for designing CARIN comes from the fact that in-car activities normally are small-scale and have short durations, which makes WiMU fail. To meet this challenge, we propose a combination-based solution that profiles all the possible activity combinations of driver and passengers in offline training and then performs recognition online. CARIN is designed to overcome the two limitations of WiMU. First, in order to handle the small signal changes caused by small in-car activities, we do not try to extract the activity of each person separately (as tried by WiMU). Instead, we try to recognize the mixed signals and treat every possible mixture as an independent pattern candidate in the classification procedure. Second, the proposed solution can recognize what activities the driver and passengers are respectively performing because all the activity combinations are profiled offline. Thanks to limited seating options in a passenger vehicle, given the fixed WiFi receiver locations, the combinations capture the signal strength of each person. Furthermore, to recognize some activities that have a sequence of body movements, like continuous head nodding due to driver fatigue, we also propose a period analysis methodology using kNN classifier to distinguish such a dangerous activity from normal (one-time) head nodding.

However, a combination-based solution is not trivial and has two serious problems: 1) combination explosion (i.e., the combination number can grow exponentially with the numbers of people and activities) and 2) activity synchronization (i.e., not all activities start at exactly the same time). First, for combination explosion, we adopt a two-step solution that first leverages equipped in-car sensors (e.g., the pressure sensor on each seat) to determine how many passengers are in the car and their locations. The number of possible combinations can be significantly reduced based on this information. In order to allow fast runtime activity recognition, we then formulate a distance minimization problem for each activity combination and further reduce the number of possible combinations. Second, for activity synchronization, our analysis of the collected data in real cars show that the durations of in-car activities are usually much shorter than those of in-room activities. Thus, this problem can be substantially mitigated by considering only some typical synchronization patterns that can then be detected using an HMM-based classifier.

Specifically, this paper makes three major contributions:

- We propose CARIN, which leverages a combination-based solution to recognize driver activities under the interference of one or more passengers, even if these activities do not start at exactly the same time. CARIN also features a period analyzer that can recognize activities with a sequence of body movements, like continuous head nodding due to driver fatigue.
- We design a novel algorithm to reduce the number of activity combinations for CARIN. We first analyze how WiFi signal changes under the interference of passengers. We then reduce the number of possible combinations by formulating a distance minimization problem and eliminate combinations with a ratio thresholding method. The computation overhead has been reduced by 72.1% on average with our algorithm.
- We evaluate CARIN in a real five-seat sedan car with more than 3,000 real-world activity traces. The results show that the average overall F1 score of driver activity recognition is 90.9% when one or more passengers are present and seated in different positions, which outperforms the state-of-the-art solutions by 32.2%.

The rest of this paper is organized as follows. Section 2 discusses the related work. Section 3 motivates our work by testing the state-of-the-art solutions under the interference of passengers. Section 4 presents the design of CARIN. Section 5 introduces the evaluation results. Section 6 concludes the paper.

2 RELATED WORK

In recent years, WiFi sensing using CSI signal has received a lot of research attention. E-eyes uses the distributions of CSI amplitudes to recognize users' daily activities with commercial WiFi devices [27]. CARM proposes a human activity recognition system based on the CSI amplitude speed model [25]. Besides CSI amplitudes, recent studies have shown that the phase of CSI signal can be used to recognize more fine-grained human activities, such as movement direction of human body [19, 30] and hand tracking [34]. Among these studies, WiDance proposes a method to remove the random noise in the CSI phase signals and uses it to detect human movement directions for interactive exergames [19]; QGesutes proposes to track human hand movements using de-noised CSI phase, and can make the tracking error below 3.56cm [34]. To further overcome the disadvantages of CSI-based activity recognition system, W. Jiang et. al. propose EI to make the activity recognition system independent of the environment changes using a deep-learning based method [11]; While those studies main focus on the recognition of a single person, the most relevant study to our work is WiMU, which can recognize the activities performed by several people with fine-grained frequency analysis of the CSI signal [23]. However, WiMU can only work on large-scale in-room activities and fails to recognize cannot tell the respective performer of each activity (e.g., the driver in our scenario). As a result, WiMU cannot work well for the in-car scenario.

Recently, there are a few studies on in-car driver activity recognition using commercial WiFi devices [3, 9, 20, 32]. Compared with IMU sensor solutions such as [10, 13], a WiFi-based solution is relatively low-cost, and does not require the user to wear devices in a specific way (e.g., smartwatch on right hand). Moreover, it can be easily be embedded into the vehicle and connected to other vehicular systems like autonomous driving controller or

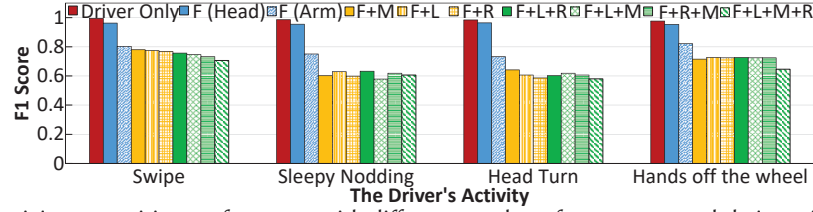


Fig. 1. The driver activity recognition performance with different number of passengers and their positions. With the front passenger's arm movement, the F1 score (the higher the better) of the driver activity drops sharply. The influence of backseat passengers does not increase when the number of them changes.

the entertainment console. Specifically, Wibot aims to distinguish head and arm movement by peak analysis and can recognize non-safety related gestures like swipe and push with an accuracy of 90.5% [20]. However, the activity set is too small and the recognition scheme cannot be expanded to a larger set of activities; WiFind keeps monitoring the CSI signal to detect whether the driver is fatigued or not using both breathing rate analysis and activity recognition with Hilbert-Huang transform [9]; WiDrive aims to improve the safety and timeliness of human driver takeover in autonomous driving [3]. ViHOT aims to track the head rotation angle for the driver, and proposes a method to minimize the head movements of other people in the vehicle using antenna theory [32]. However, the aforementioned solutions cannot be applied directly to our scenario, mainly because they cannot work well when one or more passengers are present to interfere with the driver WiFi signal. In sharp contrast, CARIN is specifically designed to recognize the activity combination under the interference of passengers.

3 MOTIVATION

In this section, we motivate the design of CARIN by testing the performance of existing driver-only recognition solution under the interference of passengers. We set the router (TP-Link Archer A7) and the laptops (ThinkPad T400 series) in a real car. The placement for these devices is the same as recommended in ViHOT [32] (as shown in Figure 2), which argues that the passengers' interference can be treated as background noise with a good device placement. To test the performance of state-of-the-art single driver recognition solutions under the passenger interference, we have implemented WiDrive [3] in this motivation test because it is shown to outperform CARM [25] and WiDance [19] in the in-car driver activity recognition scenario. The CSI signals are recorded using the modified Linux driver [7] on a packet basis, and the packet generating frequency is set to be 1000Hz. We test the performance of WiDrive for four distraction-related activities of the driver: Two arm movements - hands off the driving wheel and swiping, and two small-scale head movements - turning head and nodding off. Passengers are told to perform their movements together with the driver. To test the passenger interference, we divide the front passenger's movements into two types: head movements and arm movements. The head movement is random rotation (e.g., looking outside), and the arm movements for the front passenger are mainly chosen for vehicle-human interaction, e.g., pushing/pulling, controlling the entertainment console, and drinking/eating. For the backseat passengers, they can perform larger-scale activities like fetching items and stretching. Upon the start of driver activity, a passenger can move his/her body at any time during the driver activity, and randomly choose one in the pre-defined activity set. The passenger positions are shown in Figure 2. We mark the possible positions for the passengers as "Front (F)", "Left (L)", "Middle (M)", and "Right (R)".

We first investigate the interference caused by the front passenger. We use the metric widely used in pattern recognition, F1 score, to show the recognition performance of the existing driver activity recognition system, which is calculated as the harmonic mean of the precision and recall [6]. An F1 score reaches its best value at 1 (the perfect classification) and worst at 0. Figure 1 shows that the conclusion drawn in ViHOT [32] is partially correct because the interference of passenger can indeed be seen as noise, but only with small-scale head movements. In this case, the driver activity F1 score drops from 97.5% (Driver only) to 95.8% (front passenger making head movement). However, when the front passenger is performing a larger-scale arm movement with the driver, which causes stronger interference, the F1 score drops significantly. In this case, the average F1 score drops to

78.2%, with only four target activities. The reason for such a large drop is that when the front passenger makes arm movement, his/her arms can move out of the masking range of the WiFi antenna, as they have a larger moving range compared with the head. Thus, the CSI signal can be interfered by the passenger's arm movement and shows a difference from the driver-only pattern.

We now discuss the interference of any backseat passengers. As shown in Figure 1, the movements of backseat passengers can cause further decrease of the F1 score when they are performed together with the front passenger, especially when the driver makes only head movement: with the existence of the backseat passenger, the F1 scores of the driver's nodding off and head turn decrease by 16.3% and 14.3%, respectively. Where there is no front passenger movement, the interference of backseat passengers is found to be sufficiently small. From Figure 1, it is also interesting to see when the number of backseat passengers increases, the impact on the CSI signal does not show a corresponding increase, and the performance of the classifier does not change much. This is because the fact that each passenger has smaller movement space when the number of backseat passenger increases. Therefore, it is not necessary to tell exactly how many backseat passengers are in the car and if they are making arm or head movement. This observation can help us reduce the number of possible combinations because we only need to consider two cases: any backseat passenger movements at all or no movement in the back.

In summary, the arm movement of front passenger can have significant interference with the driver CSI signal and thus cannot be simply treated as background noise. Thus, a novel driver activity recognition solution must be designed under passenger interference.

4 DESIGN OF CARIN

In this section, we first give some preliminaries about Channel State Information (CSI), and then introduce the overview of CARIN. Afterwards, we describe each component in detail.

4.1 CSI-based Activity Recognition and Its Limitations with Multiple People

Modern WiFi devices that support 802.11n/ac usually contain multiple transmitting and receiving antennas. Multiple Input Multiple Output (MIMO) technique is utilized to increase the transmission rate and improve the link reliability. In order to perform rate adaptation or power optimization, the receiver needs to continuously monitor the situation of the wireless channel, and then reports this information back to the transmitter. This information about the wireless channel is called Channel State Information (CSI). For an Orthogonal Frequency Division Modulation (OFDM) system like WiFi, CSI consists of Channel Frequency Response (CFR) for each subcarrier between each Tx-Rx antenna pair. Specifically, under the multi-path condition, the CSI signal for a given subcarrier with frequency f at timepoint t can be represented as [25]:

$$H(f, t) = e^{-j2\pi\Delta f t} (H_s(f) + \sum_{i=1}^{N_d} a_i(f, t) e^{-j2\pi d_i(t)/\lambda}) \quad (1)$$

where $H_s(f)$ represents the summation of signals of all static paths (including Line-of-Sight signal and reflected signals from stationary objects), and second term in the parenthesis represents the summation of signals of all dynamic paths (donate as $H_d(f, t)$), which includes reflected signals from all moving objects. N_d is the number of dynamic paths. $e^{-j2\pi\Delta f t}$ is the random phase drift caused by the imperfection of WiFi devices, $a_i(f, t)$ is the complex attenuation factor and initial phase of the i^{th} path, and $e^{-j2\pi d_i(t)/\lambda}$ represents the phase change due to the propagation delay of the i^{th} path: d_i here is the length of the i^{th} path and λ is the wavelength of the WiFi signal. The changes in the length of a certain reflection path lead to the change of d_i , and the phase of the CSI signal shall be affected. As shown in Figure 2, if the changes of the dynamic path d_i are caused by human body movements (e.g., the driver), these changes in the CSI values can represent how velocities distribute for different human body parts. The activity can be inferred by investigating the velocities of different human body parts during a certain period, as the activity is a sequence of body movements. For example, we can know the driver is

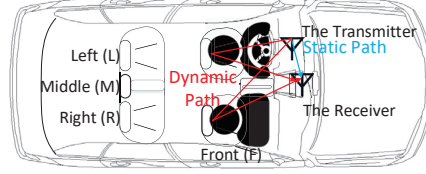


Fig. 2. An example about how CSI signal can be used to recognize activities. The reflected signal (red lines) caused by the driver contains the speed information about the driver body parts.

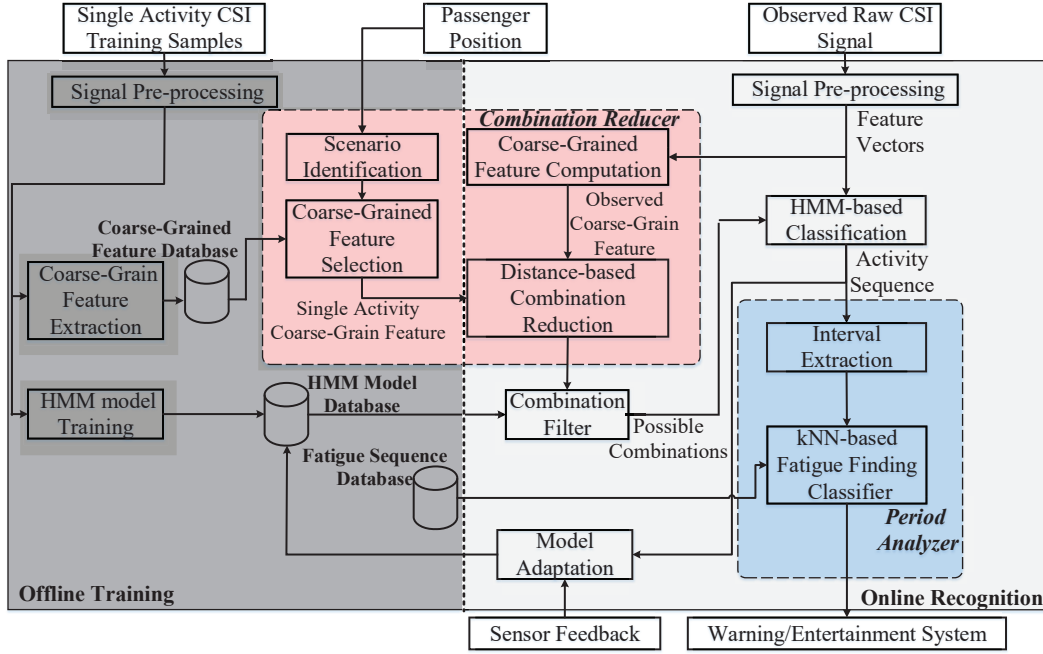


Fig. 3. Framework of CARIN.

nodding his head due to fatigue if his head moves up and down constantly, or he is eating/drinking when the speed of the arms changes in the same way as the pattern pre-trained and stored offline in the recognition system.

However, when there are multiple people in the same environment, the situation becomes more complicated. Though the CSI signal reflected by different people can be added according to Equation (1), it is difficult to get the real system response $H(f, t)$ because the interference of the WiFi signal is too strong. Thus, it is difficult to separate the CSI signal using Independence Component Analysis (ICA) or blind source separation method. Moreover, the performer for a certain activity is still unknown as the attenuation factors $a_i(f, t)$ is not known to the receiver when the activities are performed. To meet these challenges, we propose to recognize the activities based on the activity combinations instead of conducting signal separation, and reduce the computation overhead by using the coarse-grained activity features, which can be seen as the average Doppler shift power for an activity. In this way, the activity of the driver can be known when the activity combination is recognized because the position for people in the car is fixed. we also assume that $a_i(f, t)$ does not change much when an activity is performed because the wireless attenuation is mainly related to the traveling distance of the reflected signal.

4.2 Design Overview

CARIN is a WiFi-based driver activity recognition system aiming to recognize driver activities under the interference of passengers using CSI signal. CARIN consists of two parts: offline training and online recognition. For

offline training, The collected training CSI samples are denoised first, and then used as input to the coarse-grained feature extraction and HMM training component to construct the corresponding databases (i.e., coarse-grained feature database and HMM model database, respectively). The coarse-grained feature database contains the coarse-grained feature of every single activity, which shall be used in the combination reducer; The HMM model database contains the HMM model for each activity combinations, which is trained using simulated corresponding activity combination traces based on the single activity traces. Moreover, we determine the optimal time window size and overlap ratio by leveraging the method proposed by WiDrive, which can improve the performance of time-frequency analysis for recognizing small-scale activities while minimizing the computation overhead [3].

For online recognition, when the driver makes a movement, CARIN will capture the change in the CSI signal and begin to monitor the following CSI values. When the driver movement ends, the collected CSI signal is first pre-processed using conjugate multiplication proposed by WiDance [19]. After that, five major steps are carried out by CARIN: 1) Feature vector extraction: Fourier transform is applied to extract the Doppler shift power from the processed CSI signal as the feature vectors. 2) Combination reduction: Before doing the recognition, the combination reducer shall reduce the number of possible activity combinations by leveraging the offline traces stored in the coarse-grained feature database, and output all the possible combinations back to the classifier. 3) Activity recognition: The feature vectors serve as the input to the HMM-based classifier with a reduced number of activity combination candidates provided by the combination reducer. 4) Model adaption: After a single activity is recognized, the HMM models used for activity recognition are updated based on the adaption algorithm in [3]. 5) Period analysis: After the activity combination is recognized, period analysis is applied to head movements to further determine whether the activity is caused by fatigue or distraction. Figure 3 shows the framework of CARIN. Generally, CARIN has two major components: 1) Combination Reducer, and 2) Period Analyzer.

Combination Reducer: Before the classification for the activity combination, the combination reducer aims to reduce the number of possible activity combinations and the computation overhead of the HMM-based activity classifier. The combination reducer first leverages onboard sensors equipped in most cars to get the position of each person in the vehicle, and categorizes the positions of the passengers into one of the three position setup scenarios. After that, CARIN leverages the notation “coarse-grained activity feature” of an activity, and calculates a confidence score for each possible activity combination using the offline single activity coarse-grained feature from the database and the online observed coarse-grained feature. After the confidence scores for all the activity combinations are known, we eliminate those less likely ones based on the ratio thresholding rule, and send the result to the combination filter. Finally, only the selected activity combinations are sent to the HMM-based classifier to do the activity recognition.

Period Analyzer: After the HMM-based classifier recognizes a single activity, some activities need to be further analyzed in order to know whether it is caused by the reason that is dangerous to driving tasks (e.g., fatigue or distraction). Period analyzer first extracts the interval between two consecutive recognized single movements, and estimate the period and variance of the activity sequence. Then CARIN uses a kNN classifier to determine whether this sequence of activity can harm the road safety due to fatigue or distraction, and sends the final result to the central computer in the vehicle.

The main novelty of CARIN comes from three aspects. First, we propose an activity recognition system that is designed specifically for driver activity recognition under the interference of passengers. Unlike the state-of-the-art solutions that can only work for large-scale in-room activities, we leverage a combination-based approach that cannot only work with small-scale in-car activities, but also find the respective performer of each recognized activity. Second, to overcome the problem of the combination explosion, we show how the CSI model changes under the interference of passengers, and design our combination reduction algorithm based on the distance minimization model. Third, for some activity sequences (e.g., nodding off and sight off road), we propose to use the period analyzer to determine the cause of the sequence with a kNN classifier.

4.3 Combination-based Solution for Small-Scale In-Car Activity

CARIN aims to work on the small-scale activities in car under the interference of passengers using a combination-based solution. Specifically, in order to handle the WiFi signal changes caused by small-scale in-car activities, we do not try to extract the activity of each person separately (as tried by WiMU). Instead, we try to recognize the mixed signals and treat every possible mixture as an independent pattern candidate in the classifier.

Generally, a combination-based solution profiles all the possible activity combinations offline, and does activity recognition online, i.e., by expanding the number of activity patterns with the Cartesian product of existing single-person activities. Each activity combination represents a possible activity mixture, and can be treated as an independent candidate in the classification procedure. For example, if the pre-defined activity set consists of four activities with the driver and one passenger in the vehicle, there are $4 \times 4 = 16$ activity combinations. Note that if different people are performing the same set of activities, we consider it a different combination. For example, for the same activity set of "push-pull" with the driver and one passenger (A and B) in the car, we build two independent combination models in the classifier according to different performers (push by A and pull by B, push by B and pull by A). As a result, the proposed solution can recognize what activities the driver and passengers are respectively performing. Due to limited seating options in a passenger vehicle and the fixed WiFi receiver locations, the combinations capture the signal strength of each person. Though the principle of combination-based solution is simple, it is effective and can overcome two limitations of WiMU for the in-car scenario. First, the combination-based solution can identify the performer of each activity in a given combination. Second, it works better than WiMU for small-scale activities whose duration is short, because the start times of the activities can be too close for WiMU to generate a reliable frequency model [23]. In fact, the short duration is beneficial for a combination-based solution, as the synchronization issue can be alleviated (see Section 4.3.2).

4.3.1 Recognition of Small-Scale Activities. One concern is that for the in-car scenario, the scale of the activities is much smaller than the in-room counterparts due to a confined space. As a result, the frequency shift caused by the activities is not as strong as that of in-room ones and a much shorter duration compared with the in-room ones. Thus, the parameter used in the time-frequency analysis should be chosen carefully so the Doppler shift caused by the small-scale activities can be seen clearly. To solve this problem, we find the method proposed in WiDrive [3] to be effective, which leverages Short Time Fourier Transform (STFT) to capture the quick changing signal of a short-duration activity. The key feature is to overlap two consecutive time windows for increasing the visibility for time-varying signals while preserving frequency resolution. With this method, we can find the best trade off point of frequency resolution (ΔF) and time window size of STFT (ΔT), and minimize the computation overhead. In our implementation, since we are testing smaller-scale activities like head turning, we change the required minimum frequency resolution to 1.5Hz and keep the other settings suggested in [3]. Note that we apply this method to recognize an activity combination, while WiDrive tries to detect only a single activity.

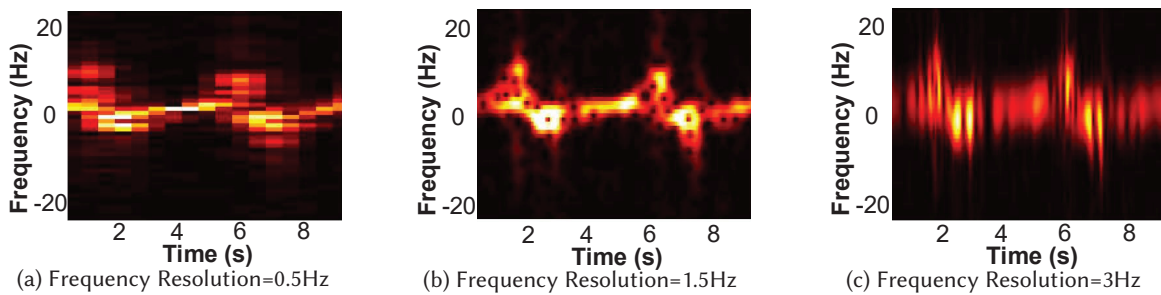


Fig. 4. Comparison of spectrums with different frequency resolutions.

Figure 4 shows the spectrums of continuous head nodding with different frequency resolutions. A 1.5Hz resolution is sufficient to show both the time-varying characteristic and correct Doppler shift in the near-zero

frequencies. Furthermore, we can see that the consequences of choosing a too small or large resolution. A frequency resolution of 0.5Hz (Figure 4(a)) needs a large time window to accumulate more data points, so the spectrum cannot reflect the time-varying characteristic of the CSI signal; A 3Hz frequency resolution (Figure 4(c)) causes almost all useful frequencies overwhelmed, due to a short time window with insufficient data points.

Another issue with small-scale activities is the detection of the start and end points of an activity combination, i.e., the segmentation problem. Though WiDrive has proposed a way to optimize the performance of Fourier transform in the recognition process, its segmentation algorithm still aims at arm movements. To detect the start point of a even smaller-scale activity like head turning and nods, we still need to find the change point in the time domain rather than frequency domain. Thus, we adopt the change point detection algorithm proposed by Wibot [20] to find the start point and end point of the activity by constructing two hypothesis: H_0 , means that there is no change point in the CSI signal, and H_1 , means there is at least one change point in the signal [20]. After that, a maximum likelihood estimation algorithm is applied to recursively find all the possible change points in the signal. To reduce the overhead of recursive computation and facilitate online processing, we use a sliding window of 0.5s in the implementation instead of using a long CSI traces: In this time window, the maximum likelihood algorithm is only called once to find the most possible change point, and it will be marked as the start point or end point based on the situation whether there is an ongoing activity in the previous time window. We choose 0.5s as the time window size because: 1) There are sufficient CSI samples in a 0.5s time window given a sampling frequency of 1000Hz. 2) As shown in Figure 5, a window size of 0.5s makes sure that there is only one change point possible so that we can avoid recursion to reduce computation overhead.

4.3.2 Synchronization among Concurrent Activities. For a combination-based solution, a major concern is the synchronization issue: The concurrent activities performed by different people can have different start points. Thus, the number of combinations could be infinite. WiMU solves this issue by finding the start point and end point for each activity in the combination, but it can only work on large-scale in-room movements. For the in-car scenario, not only the scale of the activity is usually smaller due to the space limitations, the duration of an activity is also smaller due to the fact that the driver has limited moving range and tends to finish the activity fast due to the driving task. To justify our observations, we analyze some in-room counterparts of the in-car activities in the CARM dataset [25] and compare the distribution of the activity duration. As shown in Figure 5(a), the duration for in-car activities is much smaller than their in-room counterparts, and 85% of the in-car activities last shorter than 1.6s, for a simple pulling/pushing action. For the in-room ones, only 24% of the same activities last shorter than 1.6s. The difference becomes larger when it comes to a longer activity such as fetching in Figure 5(b).

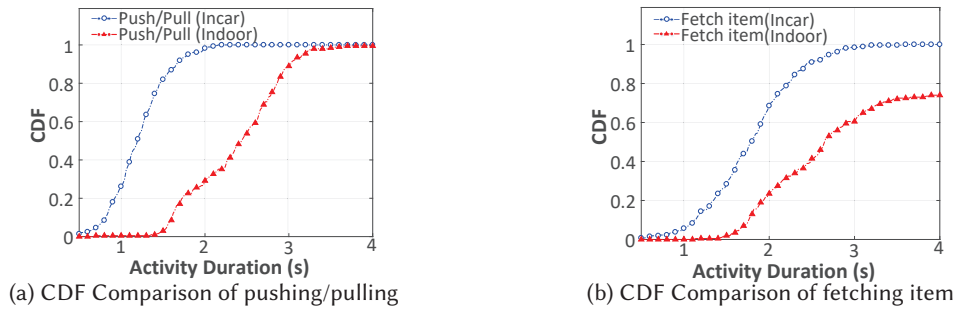


Fig. 5. Comparison of activity duration of in-car activities and their in-room counterparts. The activity duration for in-car activity is smaller than in-room ones.

With a shorter duration, the time difference among the concurrent activities in an activity combination is also small. This is because any activity that has no overlap with other activities in the combination is treated as a separate combination to be processed in another observation. Thus, for a combination with activities starting at

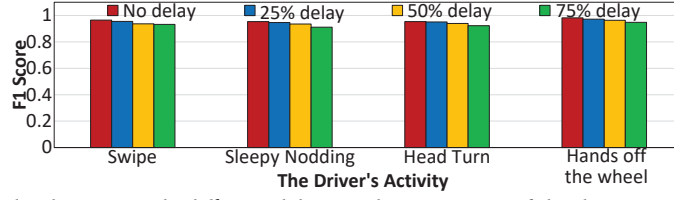


Fig. 6. The experiment result when we apply different delays to the start point of the driver activity. The F1 score only drops by 3.3% on average and by 5.5% when the delay is 75%.

different times during its short duration, its spectrum will not be significantly different from the case when all its activities start at exactly the same time. To verify our hypothesis, we test the F1 scores of the driver activity together with the front passenger's activity when the two activities start at different times with the same setup in the motivation section. To account for the time difference for different activities, we use the relative time delay (in percentage) in the test. For example, 50% delay means that the passenger starts the activity when the driver activity is halfway done. Figure 6 shows the result of the experiment. The F1 score drops only by 3.3% on average, and by 5.5% even if the delay is 75%. This degradation can be acceptable for a system (like CARIN) that favors fast runtime recognition. For systems that require a high performance, the different delays (0%, 25%, 50%, and 75%) can be modeled as different synchronization patterns, which can then be detected using an HMM-based classifier at the cost of higher computation overheads. In summary, the synchronization issue for the in-car scenario is not as significant as for the in-room scenario, due to a much smaller activity duration. For the in-car scenario, a combination-based approach can be applied to recognize the driver activity under the interference of passengers, even when the interference activities and real driver activity start at different times.

4.4 Activity Combination Reduction

A key challenge with combination-based solution is combination explosion: Because the number of possible activity combinations can grow exponentially with the increase of passengers, and quadratically with the expansion of the pre-defined activity set. The combination reducer first uses the passenger position and then leverages the coarse-grained features of the observed CSI signal to reduce the number of possible activity combinations for our HMM-based classifier. In this section, we first show how CSI changes under the interference of other activities theoretically. Then we introduce each part in the combination reducer in details.

4.4.1 CSI Changes under Interference of Other Activities. Here we first theoretically analyze how CSI changes under the interference of other activities are related to CSI signals when each single activity are performed alone, in order to provide a theoretical basis for our combination reduction algorithm in Section 4.4.3. Since up to 70% of human body is water and water is not a strong reflector to radio waves, the reflected signal strength should be much weaker than that of the original wireless signal, and the double reflection case can be neglected [34]. In Equation (1), if we omit the static component $H_s(f)$ (since it does not influence the Doppler shift), the dynamic component for a certain receiving antenna H_{d1} in the CSI signal in Equation (1) can be expressed as:

$$H_{d1}(f, t) = e^{-j2\pi\Delta f t} \left(\sum_{i=1}^{N_r} a_{ri1}(f, t)H_{ri1} + \sum_{i=1}^{N_p} a_{pi1}(f, t)H_{pi1} \right) \quad (2)$$

From Equation (2), we can see that the impact of the driver and passenger on the CSI can be added: The total response $H_{d1}(f, t)$ is the summation of the driver's impact and the passenger's impact. However, due to the noise in the WiFi signal, the true CSI values are overwhelmed in noise and often not available. To eliminate the noise term $e^{-j2\pi\Delta f t}$ while preserving the phase information in CSI, we use the method proposed by WiDance to denoise the CSI signal by conjugate multiplying the CSI signal received in two different antennas because it is shown to be effective for the in-car scenario, especially when the activity difference lies in the moving direction [19][3]. Mathematically, assuming the CSI signal received by two separate receiving antennas are $H_1(f, t)$ and

$H_2(f, t)$, respectively, then the conjugate multiplication can be expressed as:

$$\begin{aligned}
 H_1(f, t)H_2^*(f, t) &= H_{s1}H_{s2}^* \\
 &+ H_{s1}(f)^* \left(\sum_{i=1}^{N_r} a_{ri2}(f, t)H_{ri2} + \sum_{i=1}^{N_p} a_{pi2}(f, t)H_{pi2} \right) \\
 &+ H_{s2}(f)^* \left(\sum_{i=1}^{N_r} a_{ri1}(f, t)H_{ri1} + \sum_{i=1}^{N_p} a_{pi1}(f, t)H_{pi1} \right) \\
 &+ \text{Other Crossterms}
 \end{aligned} \tag{3}$$

where "Other Crossterms" contain the results derived by multiplying the dynamic part from $H_1(f, t)$ and $H_2(f, t)$, whose amplitudes are several orders weaker than that of the other terms in Equation (3) due to the strong Line-of-Sight(LoS) and other stationary reflection signals (i.e., H_s) [19]. As a result, the CSI changes for multiple activities can be viewed approximately as the summation of CSI changes with the single activity performed by each person after conjugate multiplication.

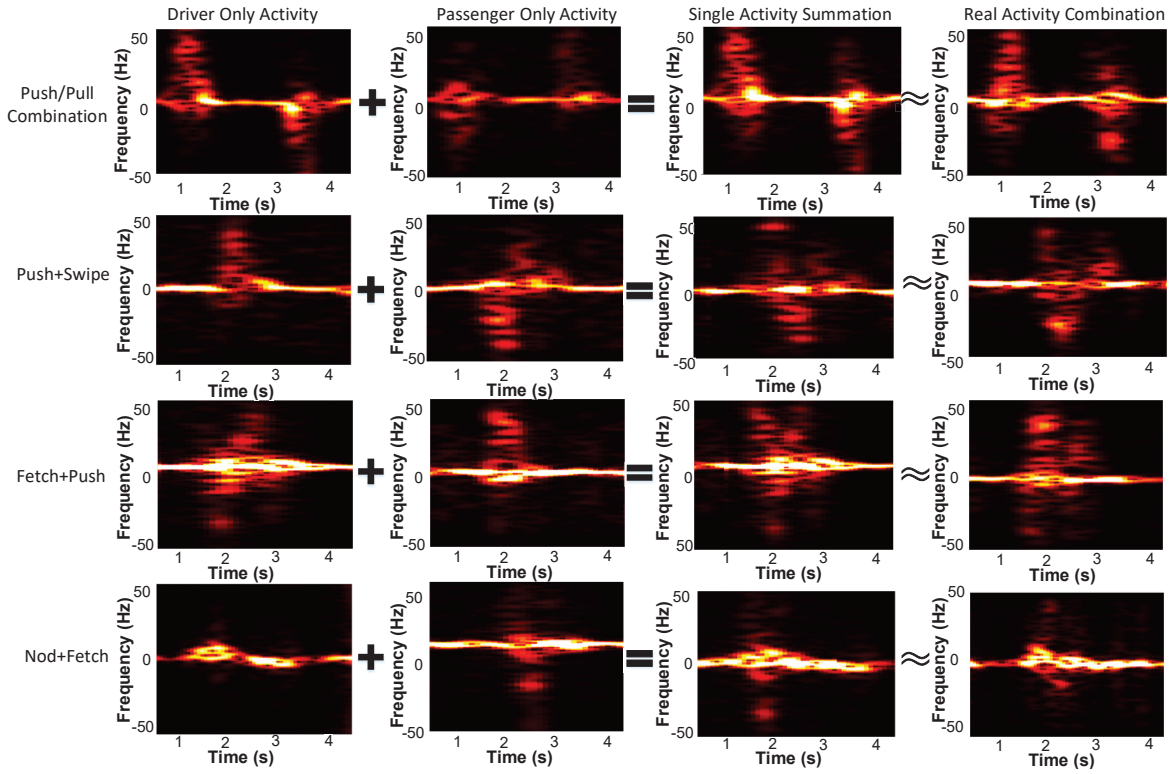


Fig. 7. The result for our validation test. We sum up the spectrums of the single activities individually performed by the passenger and driver, respectively, and then compare the summation spectrum with the spectrum of the real activity combination, where the driver and passenger perform their activities together.

To verify Equation (3), a verification test is performed in the real car with two persons. In the experiment, we first let the driver or the passenger perform the activity individually, and collect the corresponding CSI signals for these two single activities. Then the two CSI signals are transformed individually into frequency domain using STFT without summation, and the corresponding spectrums for the single activity can be obtained. Then, we let the driver and the passenger perform the same activities (as before) together, and get the spectrum for the activity combination. After that, we obtain the summation spectrum by summing up the spectrums of the two single activities (performed individually by the driver and the passenger), and compare it with the real case where the driver and the passenger act together. As shown in Figure 7, we first test a simple push and pull combination: The driver performs a push first and then a pull; accordingly, the passenger performs a pull first and then a push. Though there are small differences between the real case spectrum and the summation spectrum, the trend and the shape for the Doppler shift remain almost the same: a clear positive Doppler shift with a weak negative shift for the push/pull combination, and vice versa for the pull/push combination. Furthermore, we have tested more activity combinations like push/swipe, fetch/push, and nod/fetch and observed the same trend in Figure 7.

To quantitatively measure the spectrum similarity, we use the Distance Time Warping (DTW) to calculate the distance between the spectrums shown in Figure 7. We first calculate the upper bound of the similarity as the average distance between the spectrums of the other CSI traces from the same real activity combination to the referred spectrums (last column in Figure 7) because the distance between two spectrums from the same real activity combination is the smallest among all the activity combinations, and thus can serve as the reference distance for comparison. Then we calculate the distances from the summation spectrum, driver activity only spectrum, and passenger activity only spectrum to the referred spectrums and compare them with the reference distance. The average distance per time window for our summation spectrum is 1.54 for the selected four activity combinations (i.e., push/pull, push/swipe, fetch/push, and nod/fetch), and is only 8.7% larger than the reference distance. In comparison, the average distances from spectrums of individual activity performed by the driver and the passenger to the referred spectrum are 18.4% and 23.9% larger than the reference distance, respectively. This experiment validates our hypothesis, and we therefore use this model to design our combination reduction algorithm later in Section 4.4.3.

4.4.2 Scenario Identification. The state-of-the-art in-room solutions usually do not consider the signal difference when the user is performing the same activity at different positions because users can perform activities anywhere in a room and there shall be infinite number of possible position choices for the user. Moreover, with large in-room space, the position difference becomes weaker due to the distribution of Fresnel zone, so the difference in the CSI signal of the same activity performed in the different position is not large [37]. However, with limited space and a much smaller distance from the user to the devices, the signal difference of the same activity performed at different positions cannot be neglected for the in-car scenario. Fortunately, the position of the user in the car is usually fixed. When there are multiple people in the same vehicle, we can determine their sitting positions using onboard sensors such as pressure sensor or cameras.

However, it is still difficult to build an HMM-based classifier for every position setup, as there could be too many position setups for the same vehicle, even for a five-seat car. For example, when there is only one passenger (in addition to the driver), the passenger could sit in the front, or any of the three back seats. In order to reduce the number of possible position setups, we leverage the result from the motivation section and investigate the impact of each passenger deeply in the spectrum. For the front passenger, he/she has the largest impact on the classifier's performance, and its impact cannot be eliminated using the simple placement method. As a result, all of the arm movements should be considered by the classifier. For the backseat passengers, their impact does not increase with the number of them as stated in our motivation test due to a smaller space for movements. Thus, we consider all the backseat passenger positions as an entirety, i.e., use a binary indicator to represent the existence of backseat passengers. Furthermore, the CSI changes caused by backseat passengers are much

weaker than that of the driver and the front passenger due to the longer reflection distance and sheltering of other objects in the vehicle. As a result, the situation where only back seat passengers (in addition to the driver) exist can be eliminated. Thus, we reduce the number of different position setups to three main in-car position scenarios: 1) driver only, 2) driver with only front passenger, and 3) driver with front and any backseat passengers. As a result, we reduce the number of possible position setups from about 30 to 3, and greatly reduce the number of HMM models used in the classifier. After the current scenario is known, the result shall be used as input to the coarse-feature selection component to load the corresponding coarse-grained features for the single person activity in the database, and these selected coarse-grained features are further used to reduce the number of HMM models used by our distance minimization algorithm.

4.4.3 Combination Reduction. With the help of scenario identification, we can only load these activity combinations with the current scenario to do activity recognition. However, the number of possible activity combinations can still be large. If the activity set continues to expand, the total number of activity combinations shall grow quadratically, which is often considered infeasible in algorithm designing. Though WiMU [23] has proposed a method to reduce the number of activity combinations in the multi-people activity recognition problem, it relies heavily on the success of separating each activity in the combination, and pairing the start point with the end point for each activity. However, for the in-car scenario, the activities can hardly be separated due to the much shorter lasting duration. Thus, we propose a new method to reduce the number of possible combinations for CARIN based on confidence score for each activity combination. It uses the coarse-grained features calculated from the observed CSI signal, and our offline data of single activity as the input, and outputs all the possible activity combinations for the current CSI signal. Generally, our method contains two steps: 1) Calculate the least Euclidean distance to each activity combination, and 2) Eliminate combinations based on their confidence scores.

First, we leverage the coarse-grained activity feature defined in [23]. Mathematically, it can be calculated as:

$$\mathbf{M} = \frac{\sum_{i=1}^{(T-w)/s} \mathbf{F}_i}{(T-w)/s} \quad (4)$$

where $\mathbf{F}_i$ represents the Doppler shift power for each frequency component in a time window; w is the length for each time window and s is the stepsize used in STFT. T is the end time of the last finished activity. Generally, $\mathbf{F}_i$ can be viewed as how much energy this activity introduces in the Doppler shift in this time window, and $\mathbf{M}$ is the averaging result of $\mathbf{F}_i$ throughout the activity duration. Since the position difference cannot be neglected for the in-car scenario, $\mathbf{M}_i$ are calculated position-dependently: For example, if the activity set contains seven activities and there are three possible position for people in the car (driver, front passenger, and backseat passenger), then the total number of $\mathbf{M}_i$ is $7 \times 3 = 21$.

As shown in 4.4.1, the CSI signal for an activity combination can be seen approximately as the summation of the CSI signal for each single activity in that combination. Since the coarse-grained feature is the average version of the CSI signal, it can also be approximately added for multiple activities: If the recorded CSI trace contains N activities from N different passengers, the coarse-grained feature of this recorded CSI trace ($\mathbf{M}_{\text{obs}}$) can be approximated as the summations of N coarse-grained features of the single activities contained in the recorded CSI trace, i.e., $\sum_{i=1}^N \mathbf{M}_i$, where $\mathbf{M}_i$ is the coarse-grained activity features for the i^{th} single activity. In this summation, each $\mathbf{M}_i$ represents the coarse-grained features from different positions, and only one $\mathbf{M}_i$ for each position is allowed to appear in the summation.

However, the real $\mathbf{M}_i$ in the observed CSI value could be different from the pre-stored CSI signal of the single activity, due to the fluctuation of the transmission power and the duration variance of the activities. Thus, we assign the scaling factor a_i for each activity, and try to find the optimal a_i to each $\mathbf{M}_i$ so that the Euclidean distance between $\mathbf{M}_{\text{obs}}$ and $\sum_{i=1}^N a_i \mathbf{M}_i$ is minimized. Note that because a_i is not known a priori, the Euclidean distance for the true activity combination may be not the optimal value among all the combinations. However, the combination reducer aims only to eliminate those less likely activity combinations rather than recognizing

the combination with the one that yield the least distance. As long as the true combination is not filtered out, the result is acceptable and the final recognition shall be done in the HMM-based classifier. Mathematically, the least Euclidean distance problem we propose for each activity combination can be formulated as:

$$\begin{aligned} \min \quad & \left\| \sum_{i=1}^N a_i \mathbf{M}_i - \mathbf{M}_{\text{obs}} \right\| \\ \text{s.t.} \quad & \frac{1}{\alpha} \leq a_i \leq \alpha \end{aligned} \quad (5)$$

where α represents the allowable power fluctuation range; i represents each person in the vehicle, and N is the number of people in the vehicle (Note that backseat passengers are treated as one passenger, so $N \leq 3$). In our implementation, we set α to be 1.5 based on our observation that this value provides the best performance in terms of valid eliminations (i.e., the combinations are eliminated correctly by the combination reducer); For an activity combination, we first check each single activity in the combination, and then extract the corresponding coarse-grained feature $\mathbf{M}_i$ stored in the coarse-grained feature database. Then, gradient minimization is applied to Equation (5) to find the optimal a_i and the least Euclidean distance for this activity combination, which incurs a time overhead of only 100 μs per combination in our implementation on a 2.5 GHz Intel 7500U CPU platform. After we find the least Euclidean distance of the activity combinations to $\mathbf{M}_{\text{obs}}$, we directly assign the confidence score D as the least distance for this activity combination. This procedure is done for every activity combination in the Cartesian product, so every activity combination has its confidence score.

Then we eliminate these less likely activity combinations using ratio thresholding rule, which can be proven as the optimal trade-off method between the effectiveness and the false elimination rate [17]: First, we find the combination that has the least Euclidean distance to $\mathbf{M}_{\text{obs}}$, and define this distance as $D_{\min}$. Then, for every other activity combination, we calculate the ratio R as $R = D_{\text{other}}/D_{\min}$, where D_{other} represents the least Euclidean distance for the current activity combination. After we calculate all R values for activity combinations, we use a confidence threshold $T = 1.15$ to filter out these combinations that have a 15% larger distance than the least one because it can achieve a good trade-off between the false elimination and overhead reduction.

After all possible combinations are obtained, the HMM classifier shall only load the HMM models of these possible activity combinations, and outputs the recognition result based on the maximum likelihood for these activity combinations. Afterwards, the period analyzer shall further decide whether this activity belongs to a sequence of head movements, and try to infer the cause of the sequence with the well-trained prediction model.

4.5 HMM Model Adaptation

For in-room activity recognition, the HMM classifier is often trained offline, and its parameters do not change after training. However, for the in-car scenario, only relying on the offline trained HMM classifier may not work well when it is applied to a different vehicle or driver, or when a driver changes his activity pattern. Thus, in order to adapt the HMM classifier to a new environment or new driver, we adopt a simple adaptation algorithm proposed in WiDrive [3] for each activity combination. Once the performed activity combination finishes and can be confirmed by other add-on sensors (e.g., camera) or a human helper, the HMM parameters for that activity combination are adapted to fit the activity pattern of the new environment or driver. Specifically, WiDrive uses the Expectation-Maximization (EM) algorithm to adapt the HMM models in the classifier. Upon the arrival of a new CSI sample, a weight term W is assigned with it, and the cost function $Q(\theta, \theta_i)$ is rewritten as:

$$Q(\theta, \theta_i) = \sum_{t=1}^{T-1} \sum_S p(S|O_t, \theta_i) \log p(O_t, S|\theta) + W \sum_S p(S|O_T, \theta_i) \log p(O_T, S|\theta) \quad (6)$$

where θ represents the HMM parameters for the activity combination to be updated; θ_i represents the current HMM parameters for that combination; O_t is the t^{th} old CSI sample; S is the hidden state set; T is the number of old samples. By taking the partial derivation of θ_i , the update equation for the HMM parameters can be derived

as a function of the current CSI sample O_T and the adaptation result derived from the previous CSI sample (i.e., the result from the last adaptation round). After the adaptation for the current CSI sample is done, O_T is treated as an old sample and stored in the system with the current calculation results from the EM algorithm to prepare for possible future adaptations. With sufficient CSI traces, each HMM model in the classifier can be adapted to a new environment. The detailed design and analysis of the adaptation algorithm are available in [3].

4.6 Period Analysis

Usually, a single arm movement can be a sign of dangerous driving behavior because it is directly related to the road safety (e.g., hands off the driving wheel). However, for the head movement, the issue is more complicated because the sign is usually not only the single movement but a series of continuous movements. For distraction and fatigue driving, the most well-known issue is sight off road (e.g., looking outside, texting) and nodding offs, respectively [1][9]. To identify those two activities, we further design a component called period analyzer. While distracted head turning (i.e., sight off the road) can be solved by using a timer when the first head turning is recognized and setting a certain threshold, nodding off is more complicated because a single nod is meaningless and we need to infer fatigue by monitoring several nod activities during a certain period. Based on the statistics of the activity sequence (e.g., period, variance, etc.), we could infer the cause for the sequence with the well-trained prediction model. In this part, we first introduce how to extract the statistics of the activity sequence, and then we discuss how to generate the model for deciding the cause of continuous nod sequence.

The most used technique to estimate the period of a signal is auto-correlation function with peak analysis [39]. However, auto-correlation is easy to be interfered, especially for multi-people scenario where the interference from the passenger is much stronger than background noise. Moreover, auto-correlation function incurs heavy computation (multiply-accumulation) and is not suitable for the in-car scenario. Since a single activity can be recognized in our case, after recognizing the first nod activity, the following CSI shall be monitored continuously. In this period of time, all the start points for the driver head movement are recorded, and the interval between two continuous head movements is calculated and stored. When the monitor time is up, we extract the mean value and the standard variation of the list of activity intervals, which are used as input to the fatigue finding classifier. Though the single activity could be recognized wrong and the estimated period and variance can also be inaccurate, it is still better than using auto-correlation and is shown to have good performance as long as the monitoring duration is long enough (see Section 5.6)).

Then we build our fatigue classifier with the input of the period and standard variation for the head movement sequence. We first record more than 60 traces of nodding off and 60 traces of random head movements (e.g., neck relaxation, checking rear-view mirrors, scratch the head, etc.) in a real car. From the traces, we find two key observations: 1) a fatigued nod usually lasts about 3s to 5s, so the mean of the time between the two peaks should be in that range; 2) As an unconscious movement, the nod behavior should be nearly periodic. Thus, the variance for it should be small. Figure 8(a) shows the spectrum of a typical nodding off, and we can find that its period is about 4s and the variance of the nodding is 1.32. Figure 8(b) shows the distributions of nodding off and random nods, where nodding offs are marked with blue dots. We can see that different nodding off traces often have similar mean values and variance due to the sub-consciousness status of the driver, while random head motions are sparsely distributed in Figure 8(b).

To find the classifier, we first try the variation threshold-based method in Smokey [39]. However, we find that this method cannot fit our data well due to the fact that there is not a certain pattern in the nod sequence like smoking, and the period of nodding off varies more significantly than smoking activity. Another method is to use the coefficient of variance (cv), which can be seen as a linear discriminator (purple line shown in Fig 8(b)). However, due to the non-linear boundary, the performance of cv threshold is not good as well.

To solve this problem, we use a k Nearest Neighbor (kNN) classifier for this task. kNN is widely used in text categorization and image recognition [33][38]. Generally, a kNN classifier stores all the training data in the memory. When the testing sample comes, the classifier finds a group of k samples in the dataset that are

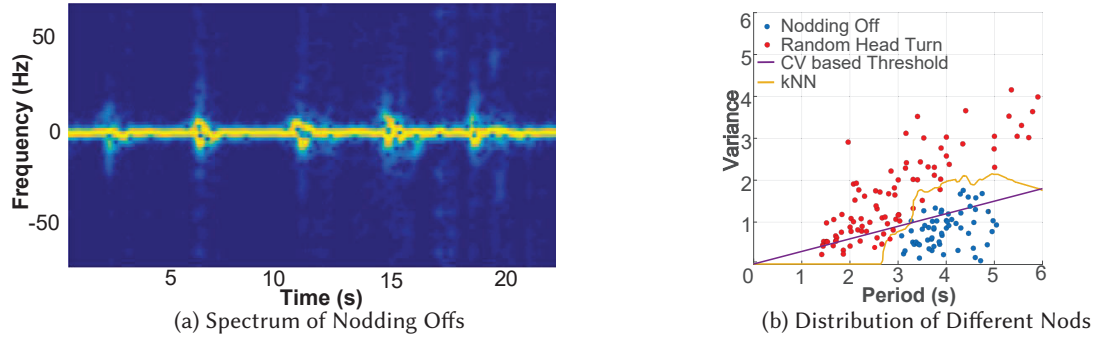


Fig. 8. The Spectrum of nodding off and the distributions of different nodding activities where blue dots are nodding off.

nearest to the unknown sample (e.g., based on distance functions). From these k samples, the label (class) of the unknown sample is determined by the majority of the label in the k data points. We choose kNN classifier for several reasons: 1) kNN can be used in the non-linear case and performs well on the low dimension data [5], which is exactly our scenario shown in Figure 8(b). 2) Since kNN relies on the nearest neighbors voting to do the classification, the effect of outliers in the training data is negligible as its distance to the sample shall be large. Moreover, the new collected data can be directly included in the training data set so the kNN can be adaptive. We iteratively run cross validation test with k varying from 1 to 20, and choose $k = 7$ as it yields the best classification performance.

However, a kNN classifier is well-known to have the high computation and storage overheads. In order to find the k nearest neighbors, the classifier first calculates all the distances from the testing sample to each data point stored in the dataset, then compares these distances and finds these k points. Its computation complexity is $O(nkd)$, where n is the size of the pre-stored dataset and d is the dimension of the data, and a storage space of $O(nd)$. Fortunately, the reduction of computation overhead in kNN has been well studied. After applying some well-known kNN algorithms, like k-D trees [4], the computation overhead becomes acceptably small compared with the long monitoring time of the period analyzer. Similarly, for storage space, the required storage space for a dataset containing 1024 two-dimensional 32-bit float vectors (10 times larger than our dataset) is about $4 \times 2 \times 1024 = 8\text{KB}$, which is also acceptable for the onboard embedded computer in our in-car scenario. With the reduced kNN computation overhead, CARIN can efficiently recognize a nodding off sequence with an F1 score higher than 97% and a false positive rate lower than 2% in our evaluation (see Section 5.6)).

4.7 Discussion

One concern for CARIN is how to detect an activity that is not included in the activity set, i.e., the “unknown” class. For an HMM classifier, there are two typical methods to handle this issue. First, we can collect several most performed unknown activities, and train a corresponding HMM model for the “unknown” class. However, this solution lacks the ability for generalization since we cannot include all unknown activities of a driver in training. Second, we can adopt the filler model [15], which can be seen as an averaging version of the all HMM models in the classifier. If the sample is a pre-defined activity, the likelihood of the corresponding HMM model should be larger than the averaging HMM model (i.e., the filler HMM model), and vice versa. If the likelihood of the filler model is larger than all other HMM models of activity combination, this sample is considered as an “unknown” activity. As the combination set expands, the filler model can also be expanded to include more activities performed by the passenger as well. To test the influence of unknown activities performed by the front passenger with the filler model, we conduct an experiment with several new activities (e.g., adjusting shading board, stretching himself/herself) performed by the passenger. The influence of unknown activities on the HMM-based classifier is relatively small, and the overall F1 score for the recognition accuracy for the driver drops by 4.7%. Thus, we use the filler model to filter the unknown activities in our HMM classifier.

Another concern is that the different inner layout for different vehicle types. Though we design and evaluate CARIN mainly for a five-seat sedan car because it is the most popular vehicle type on the road, CARIN can also be applied into some other vehicle types, such as small SUVs or trucks because their internal layout is similar to a five-seat car. For passenger vehicles with different numbers of seats (e.g., an eight-seat Minivan or SUV), we can expand the number of combinations to consider those extra seats at the cost of higher computation overheads. For commercial vehicles with even more seats (e.g., bus), though they are not the focus of our paper, we believe some special designs shall be necessary to account for the effective range of CSI-based recognition system in such a large vehicle room, which can be our future work.

Moreover, WiFi-based solution can also face the problem like segmentation on or during a continuous movement, where finding the start point for a certain activity can be different from the case with no movement at all in the vehicle. In fact, segmenting an activity from a set of continuously performed activities by any given user is a challenging problem in all WiFi-based solutions [23]. However, we have already shown that the duration of the in-car activity is much shorter than its in-room counterpart in Section 4.3.2. Moreover, in case that the passenger does perform a continuous movement, CARIN can monitor the duration of it with a timer. Once the movement duration becomes longer than a threshold, CARIN can treat this CSI signal as background noise, and dynamically change the probability models (e.g., probability distribution functions for H_0 and H_1) used in segmentation algorithm in Section 4.3.1 to adapt to the continuous movement. For instance, the probability models used by H_0 and H_1 shall be assigned larger mean and variation values because the “background noise” (background noise plus the continuous movement) becomes stronger, and the probability threshold of detecting a start point in the time window is also higher to decrease the false alarms caused by the continuous movement.

5 EVALUATION

In this section, we evaluate the performance of CARIN using CSI activity traces in a real car. We first introduce the experiment setup in Sections 5.1. We then compare CARIN with other well-designed baselines (Sections 5.2 to 5.9). At last, we test the overhead of CARIN (Section 5.10).

5.1 Experiment Setup

5.1.1 Testbed Setup. The testbed consists of one commercial WiFi router as the transmitter, and one laptop (the receiver) equipped with wireless Network Interface Cards (NIC). We use TP-Link Archer A7 as the transmitter, and a ThinkPad T400 with Intel 5300 wireless NICs (chosen to fit the modified driver [7]) to establish the communication link in the car. The WiFi link is set up to work on channel 165 at 5.825GHz. We generate packets using the *ping* command at a transmission rate of 1000pkt/s, and CSI values are collected with the modified network driver on a packet basis [7]. The placement of our devices is shown in Figure 9. The transmitter is placed in the middle of the dash board and the receiver lays between the driving wheel and the panel. In a real product, the receiver and the transmitter can be implemented with embedded systems so that they can be installed into the car body easily. After we obtain the data in the receiver, we process the data with the simulation tool (MATLAB) on a desktop in our lab with a CPU of Core-i7 7500U and 16GB RAM.

5.1.2 Candidate Activities Collection. For people in each possible sitting position in the car, the selected activities for each person are listed as follows:

- **Driver Activity:** We choose driver activities mainly based on safety purposes, and the major factors with driving safety would be distraction. Thus, we choose several distraction-related activities, such as fetching items (**F**), drinking/eating (**D**), turning head back (**T**), and single nodding (**N**). To be more general, we also include some previously studied activities like pushing (**Push**), pulling (**Pull**), and swiping (**S**), which could be used for interacting with the vehicle in the future [20][3].
- **Front Passenger Activity:** The front passenger has the most significant impact on the CSI signal other than the driver. In the real world, the front passenger focuses mainly on the human/vehicle interactions (such as turning up the radio volume or AC’s temperature). Thus, we choose pushing (**Push**), pulling (**Pull**),

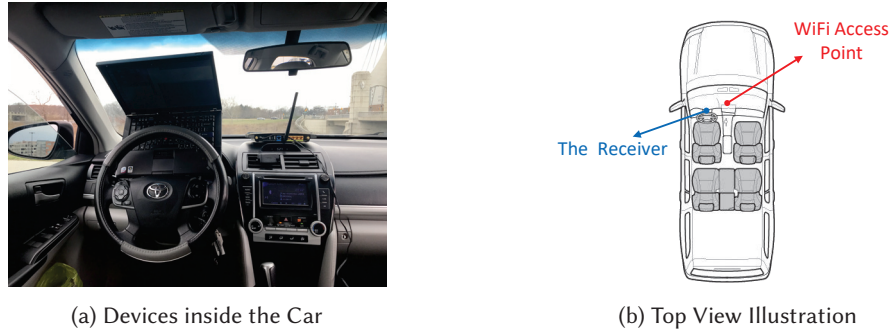


Fig. 9. Device placement inside the car.

and swiping (**Swipe**) for the front passenger as the ways to interact with the vehicle. We also include more interfering activities such as fetching gearbox, drink/eating and adjusting window in section 5.8.

- **Backseat Passenger Activity:** Compared with the driver and the front passenger, the backseat passenger has a smaller impact on the CSI signal, as shown in Section 3. In the following evaluation, we still select pushing and pulling for general purposes, and a typical larger-scale activity (i.e., fetching the door on the opposite side) to stress the recognition performance of CARIN.

Each person can stay still (“Empty (E)”) while others perform an activity. Thus, there are eight driver activities, four front passenger activities, and four back passenger activities in total for the single activity set. We also include the class of “unknown” activity (U) using the filler model in [15].

In CARIN, each activity combination should have its corresponding HMM model, but such a large number of activity combinations requires a large amount of activity traces to be collected in the training phase. We use our theoretical model in Section 4.4.1 to solve this problem. Specifically, since the CSI signal is addable, we sum the CSI training samples of each single activity contained in the combination to simulate the CSI trace in the multi-people scenario, e.g., a training sample of “push-pull” combination is produced by adding a “push” only CSI sample with a “pull” only sample. To simulate these samples, we first collect more than 50 traces of each single-person activity per user from five volunteers (4 males and 1 female) in a real car, then generate more than 200 simulated samples for HMM training per activity combination. After that, we apply the well-known Baum-Welch algorithm to train the HMM models offline [28]. These single activity traces are also stored offline in the coarse-grained feature database used by combination reducer. After that, we collect 40 traces for each activity combination per volunteer as our test set. For each activity combination, the driver starts the movement with a voice prompt. After that, the passengers can start performing activities at any time while the driver is still moving. For the same driver, passengers are randomly selected before each time the driver performs the activity. Though all traces are collected in the parking lot, previous studies have shown that the impact of the outside objects is small compared to the activities in the vehicle so the mobility issue can be neglected [9][3]. Note that we have a small-scale driving test (see Section 5.9) to test the environment changes in a moving car.

5.1.3 Passenger Position Setup. In order to test the robustness of CARIN in terms of different position setups, we use four different setups as shown in Figure 10. Figure 10(a) corresponds to the front passenger scenario, and Figure 10(b) and (c) corresponds to the front and back passenger scenario. Though the impact is small when there is only one backseat passenger (i.e., Figure 10(d)) in the vehicle, we still test the performance of CARIN in order to prove the correctness of our categorization on the different passenger position setups.

5.1.4 Baselines. For comparison, we choose baselines as follows:

- **WiMU:** WiMU is a state-of-the-art solution for in-room activity recognition where multiple people perform activities together [23]. It can find the activity combinations performed by a group but cannot associate

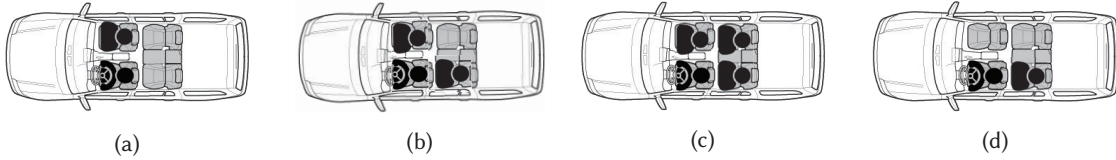


Fig. 10. Top view of different position setups.

each activity with its performer. Here we extend WiMU by checking the similarity for the single activity model, then assign the one with the largest similarity score as the driver activity. However, WiMU is not designed for in-car activities whose scale is much smaller than in-room ones. Moreover, it also does not consider the CSI difference caused by the position difference. As a result, when the position for passenger changes, the F1 score of WiMU drops sharply as well.

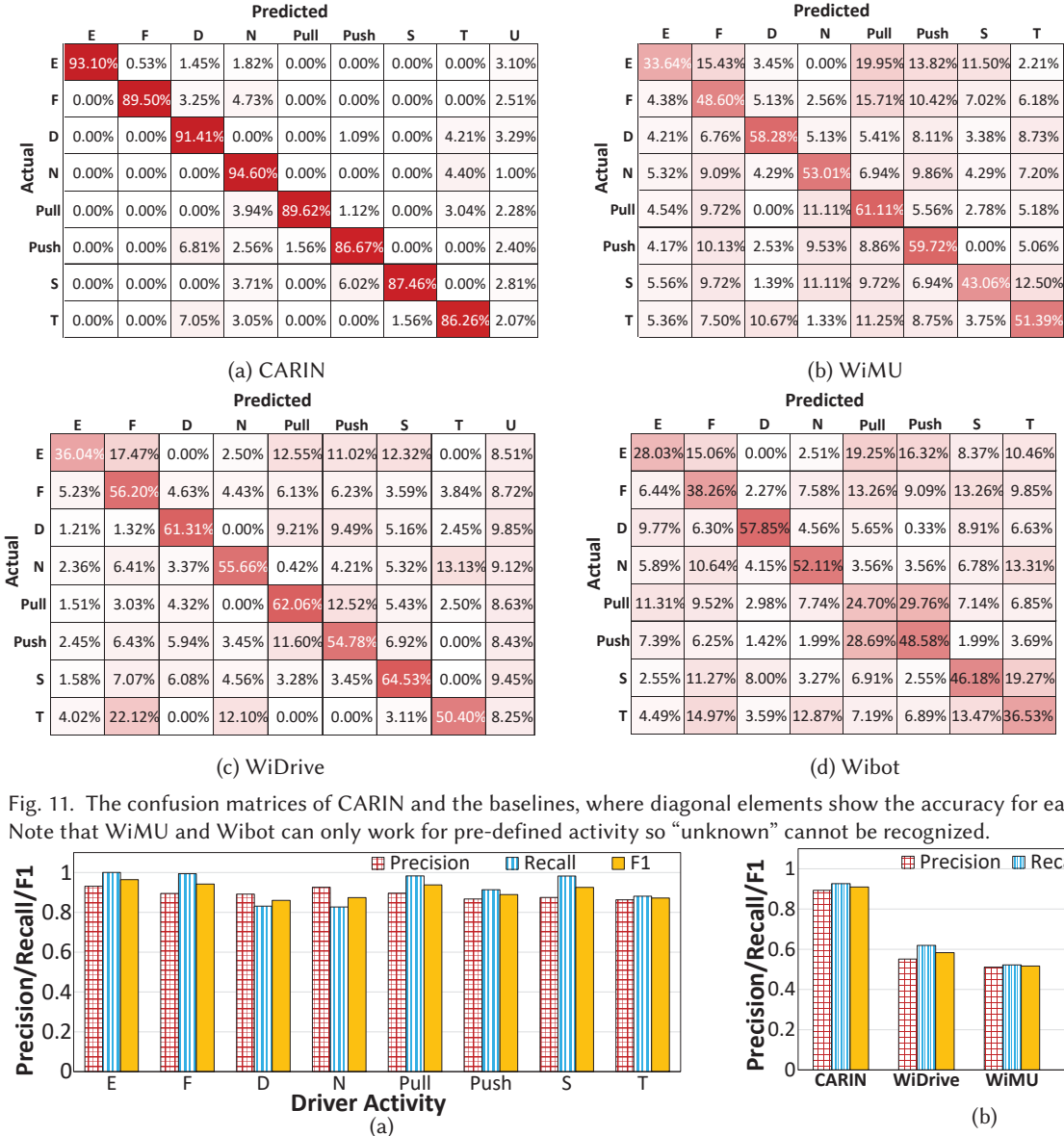
- **WiDrive:** WiDrive is a real-time in-car activity recognition system for takeover preparations [3]. It optimizes the parameters used in time-frequency analysis, such that frequency shift caused by the activities can be recognized. Moreover, its classifier is more robust than that of WiMU: First, WiDrive also uses an HMM-based classifier while WiMU uses Jarccard index. Second, WiDrive's HMM model is trained with all the activity samples of the driver, no matter what activity the passenger performs. Thus, WiDrive is capable to work under the interference of passengers to some degree in the vehicle. However, it is still designed for driver only, and its recognition performance drops when there are multiple people in the vehicle.
- **Wibot:** Wibot is also a driver-only activity recognition scheme, which relies on the number and locations of peaks in CSI phase along with the phase difference between two adjacent subcarriers to do recognition. It uses the same pre-processing algorithm of WiDrive, but its classifier is less robust than that of WiDrive. Moreover, Wibot aims mainly for distinguishing drivers' arm movements and head movements, which are much more coarse-grained than those activities considered by WiMU and WiDrive.

5.2 Performance of In-car Activity Recognition

Here we evaluate the recognition performance of CARIN and the baselines with our data set for the scenario where the driver is with the front passenger. We first show the confusion matrices for CARIN and the baselines. After that, we show the F1 scores for the activity combinations for CARIN and WiMU.

Figure 11 shows confusion matrices for CARIN and baselines, whose elements on the diagonal show the accuracy for each activity. The accuracy of CARIN is 89.2% on average in this front passenger only scenario, which is 37.2% higher than that of WiMU and 22.2% higher than that of WiDrive. The key reason for WiMU's inferior recognition performance is that it is not designed for in-car activities, and cannot find the start point and end point for each activity in the CSI signal. As a result, WiMU cannot work properly to determine the possible activity combinations and generate corresponding virtual samples. Moreover, the classifier used in WiMU is much simpler than that of CARIN, making its recognition performance worse when it comes to in-car environment. As for WiDrive, its recognition performance is slightly better than that of WiMU because WiDrive is designed for small-scale in-car activities. However, with the interference from the front passenger's activity, its overall accuracy drops to 55.3%. Unlike WiMU that shows a uniform drop on accuracy, some activities show much larger drops than others in WiDrive, mainly due to the difference in the scale of activities, such as turning head and nodding. The third baseline, Wibot, performs the worst among the baselines because peak analysis is less robust than frequency analysis, and prone to be interfered with noises and passengers' activities. Moreover, unlike the HMM-based classifier used in CARIN, the kNN classifier used in Wibot cannot reflect the time-varying characteristics of an activity. CARIN is not only designed specifically for the small-scale in-car activity, but also uses the combination-based solution to increase the recognition accuracy under the interference of passengers in the same car. As a result, CARIN achieves the highest accuracy among all the schemes.

Besides recognition accuracy, Figure 12(a) shows the precision, recall, and F1 score of CARIN for each activity: Precision represents how many samples are classified correctly, and it is the same as accuracy for a multi-class



classification problem; Recall shows how many samples are classified correctly in the same recognition result set. The overall precision and recall of CARIN are 89.2% and 92.6%, respectively, which indicate that CARIN can achieve a good trade-off between mis-recognition and false alarms. As shown in Figure 12(b), CARIN outperforms WiDrive by 33.4% and 30.1% in terms of precision and recall, respectively, and outperforms WiMU by 37.6% and 39.2% in terms of precision and recall, respectively. As a result, the overall F1 score of CARIN is higher than that of WiDrive by 32.2% and even more for other baselines.

		Driver Activity							
Passenger Activity		E	F	D	N	Pull	Push	S	T
	E	97.59%	91.75%	82.91%	91.83%	97.81%	97.93%	85.42%	85.38%
	Pull	76.45%	86.82%	77.38%	83.57%	77.32%	88.72%	84.15%	70.61%
	Push	89.50%	87.17%	78.71%	74.69%	84.37%	86.47%	88.46%	83.29%
	Swipe	87.49%	79.42%	69.13%	76.44%	71.84%	80.23%	85.30%	69.94%

(a) CARIN

		Driver Activity							
Passenger Activity		E	F	D	N	Pull	Push	S	T
	E	35.20%	26.21%	20.32%	19.27%	37.21%	30.23%	23.21%	18.23%
	Pull	21.21%	12.40%	12.13%	12.46%	17.12%	19.21%	14.35%	10.32%
	Push	11.25%	9.21%	9.78%	9.11%	19.45%	10.31%	16.21%	7.21%
	Swipe	16.21%	21.42%	15.27%	15.31%	13.88%	11.65%	7.81%	7.89%

(b) WiMU

Fig. 13. The driver activity recognition performance (F1 score) for each activity combination of CARIN and WiMU, where each element in the matrix shows the F1 score for the corresponding activity combination.

Since the front passenger can also interact with the vehicle with his/her movements, oftentimes his/her activities also need to be recognized by the vehicle together with the driver activity. Here we evaluate the recognition performance of CARIN in terms of F1 score for each activity combination. The F1 score of the activity combination is calculated as the F1 score for recognizing not only the driver activity correctly, but the passenger activities, i.e., a CSI sample is recognized correctly only if all the involved activities are recognized correctly. Here we only compare CARIN with WiMU because WiMU is the only baseline that can recognize activity combinations. Figure 13 shows the results of CARIN and WiMU. We can see that the average F1 score for CARIN is still above 83% while the average F1 score of WiMU is only 16.5%. Thus, WiMU cannot be implemented in a vehicle to help the front passenger for human-vehicle interactions while the driver is performing an activity. On the other side, CARIN works well with the selected 32 combinations, and achieves a good recognition performance among these combinations. The main reason for CARIN's superior recognition performance is that it is specially designed for the in-car environment, while WiMU cannot work well for small-scale in-car activities with short durations. This result demonstrates that CARIN cannot only work under the interference of passengers, but also can recognize the activities performed by the passengers in need of interacting with the vehicle if necessary.

5.3 Recognition Performance with Different Position Setups

Here we show the F1 score of CARIN with different position setups in Figure 10 to justify CARIN of its scenario identification component. We present the F1 score of each activity for CARIN, and compare it with the baselines.

Figure 14 shows the comparison of CARIN and the three baselines in terms of F1 score in a five-seat car. Wibot performs the worst for setups (a), (b), and (c) as its classifier is not robust, and designed for a small-size activity set with only five activities. Thus, Wibot is prone to be interfered with the passenger's activity. WiMU achieves the F1 score of 50.6% and 43.4% for setups (a) and (b), respectively. WiMU also fails to consider the position difference. As shown in Figure 14(d), the overall F1 score is only 26.8%, which is much lower than that of Wibot and almost the same as random guess. The reason for this decrease is that WiMU considers setups (b) and (d) the same because they both have two persons, but they are counted as two different position scenarios in CARIN. WiDrive performs better than WiMU, and has an overall F1 score of 58.4%, which is 7.8% higher than that of WiMU. This is mainly because that WiDrive is designed for in-car activity, and it does have some robustness under the interference of passengers because all samples for one activity (no matter what activities the passengers perform) are included in the training of the classifier. WiDrive and CARIN achieve similar recognition performance for position setup (d) because CARIN does not consider the case where there are only backseat passengers, so CARIN performs similarly with WiDrive in this case. However, WiDrive's recognition performance is still lower compared with CARIN for other position setups: The overall F1 score of CARIN is 31.1% and 29.8% higher than that of WiDrive, respectively. There are two main reasons for CARIN's superior recognition performance. First, CARIN considers the difference on the passenger activity, and trains the classifier based on each activity combination. Second, it eliminates many unlikely combinations using the combination reducer, which not only reduces the computation overhead, but also improves the recognition performance.

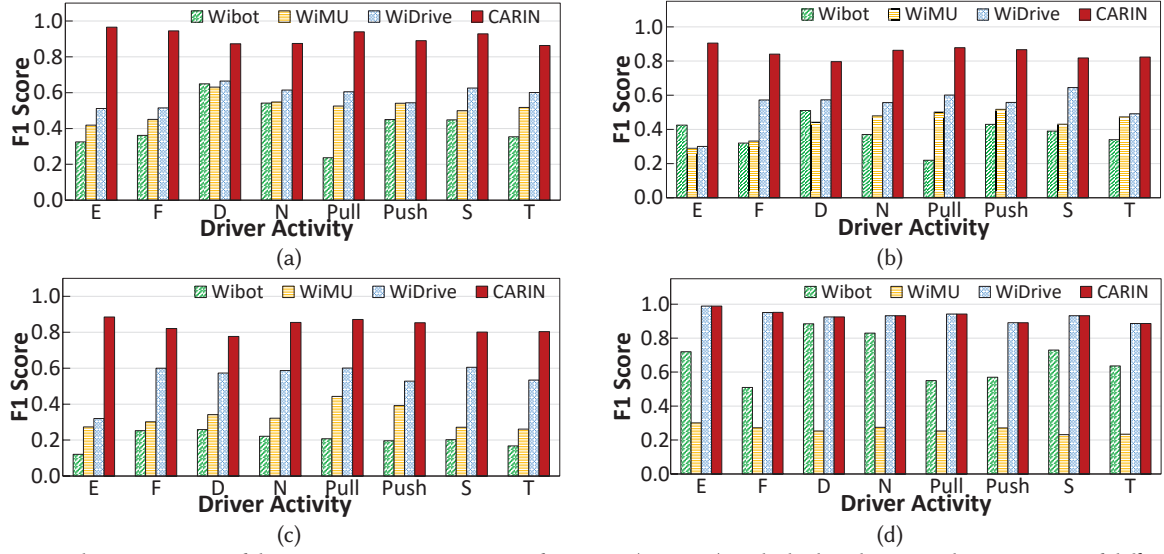


Fig. 14. The comparison of driver activity recognition performance (F1 score) with the baselines per driver activity of different position setups. Each sub-figure corresponds to the position setup shown in Figure 10.

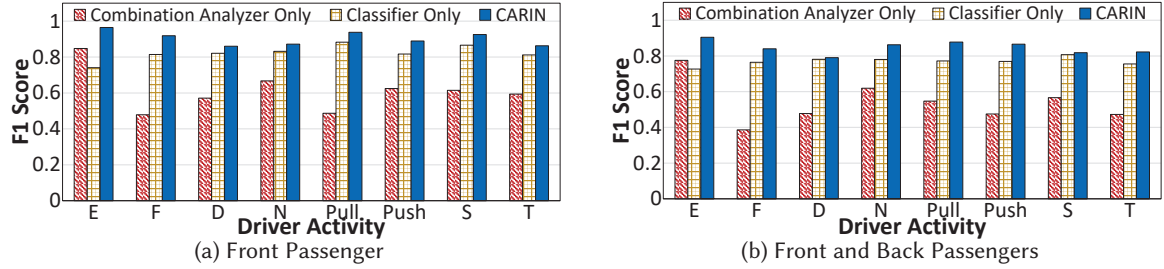


Fig. 15. Comparison of CARIN with its variants.

5.4 Effectiveness of CARIN's Components

Here we show the effectiveness of each component in CARIN in terms of F1 score by comparing CARIN with two baselines: 1) Combination Reducer Only: We only rely on the combination reducer to recognize the activity, i.e., the combination whose coarse-grained feature summation yields the minimum distance to that of the observed CSI signal is chosen directly as the final recognition result, in order to see whether the combination reducer only is sufficient for activity recognition. 2) Classifier Only: We disable the combination reducer, and all activity combinations are considered in the HMM-based activity recognizer.

From Figure 15, we can see the necessities and the recognition performance improvements for each component of CARIN. Combination Reducer Only can achieve an overall F1 score of 61.2% and 53.9% for the front passenger scenario and front and back passenger scenario, respectively. This is because the combination reducer aims for combination elimination, and relies only on the coarse-grained activity features to do the task. The coarse-grained feature cannot reflect how the activity changes during the time, so it is not sufficient for activity recognition. The only exception is the empty activity, where the driver performs no activity. In this case, the coarse-grained features can be seen as the feature for the passenger activities only given a small activity set and weaker reflected signal than the driver. On the other hand, CARIN also outperforms Classifier Only by 8.1% and 7.5% for the front passenger scenario and front and back passenger scenario, respectively. From this comparison, we can see that besides computation overhead reduction, the combination reducer also improves the recognition performance

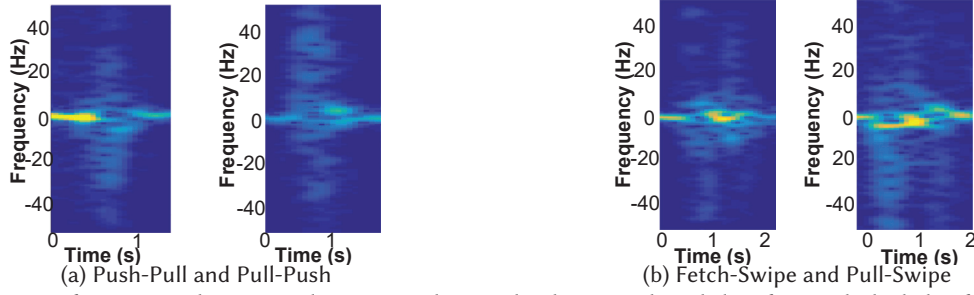


Fig. 16. Two set of CSI traces that are easily recognized wrong by the HMM-based classifier. With the help of combination reducer, one of them can be eliminated before the classification step so the recognition performance is improved.

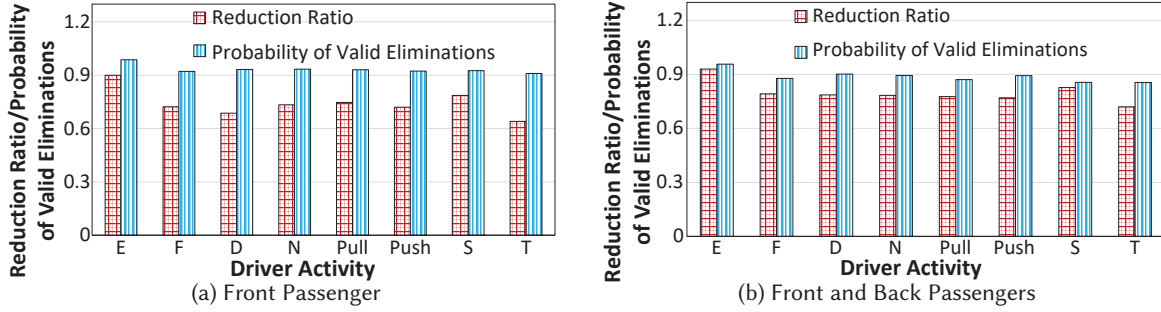


Fig. 17. Computation reduction ratio and probability of valid eliminations for each activity. CARIN manages to reduce 72.1% computational overhead with 93.5% correct eliminations on average.

of our classifier. This is because with the reduced number of combination candidates, the robustness for each remaining class in the classifier increases. As a result, the sample with a larger distortion can be tolerated in the classifier and be classified to the right class. Figure 16 shows two groups of activity combinations that are easily recognized wrong by Classifier Only. Due to the same frequency shift trend, Classifier Only cannot achieve good recognition performance for these combinations. Fortunately, the combination reducer can find their difference using the coarse-grained feature: The two CSI traces in Figure 16(a) shows the difference frequency power in positive and negative lobes, while the two CSI traces in Figure 16(b) show frequency power difference: The fetch-swipe combination incurs less Doppler power in the frequency domain than that of pull-swipe combination. This difference is reflected in the coarse-grained feature as the average version of Doppler shift.

5.5 Performance of Combination Reducer

Here we show the effectiveness and correctness of the combination reducer in terms of computation overhead reduction. We first evaluate the performance of the combination reducer by showing how much computation overhead it can reduce, and then we show how to select the optimal parameters (i.e., the confidence threshold and the allowable power variant range) in the combination reducer.

Here we use *reduction ratio* to measure the overhead reduction performance of CARIN, which is calculated as the ratio of computation time of CARIN over the baseline Classifier Only. We use the definition of *valid elimination* to represent the case where the true activity combination for a recorded CSI signal is not filtered out by the combination reducer. We use the probability of valid eliminations to measure the correctness of the combination reducer. Figures 17(a) and (b) shows the overhead reduction ratio and the probability of valid eliminations for each activity for the front passenger only scenario and front and back passenger scenario, respectively. We can see that the computation overheads are reduced by 72.1% on average in Figures 17(a), and even more in Figure 17(b). The chance of false elimination is also small for CARIN: The probability of valid eliminations is 93.5% and

90.4% in Figure 17(a) and (b), respectively. Though these results indicate that the accuracy of CARIN cannot exceed these values, compared with the case where the combination reducer is disabled, CARIN still increases the recognition performance overall by filtering out those unlikely combinations.

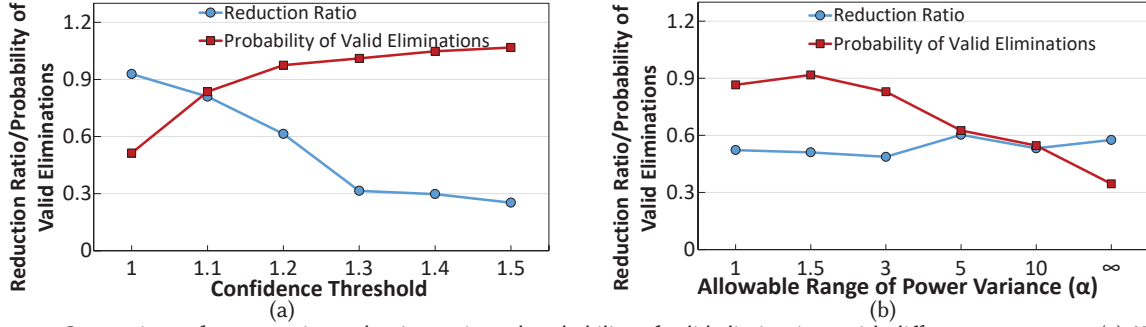


Fig. 18. Comparison of computation reduction ratio and probability of valid eliminations with different parameters. (a): With different confidence threshold. (b): Constraints on the power variation.

Figure 18 shows the performance of the combination reducer with different confidence thresholds (i.e., the acceptance range for the ratio thresholding rule) and the allowable range of power variance (i.e., α in Equation (5)). From Figure 18(a), we can see that the trade-off between the reduction ratio and probability of valid eliminations: With a lower confidence threshold, the number of combinations that are within the confidence threshold becomes smaller. With a smaller number of possible combinations in the HMM-based classifier, the computation overhead becomes smaller but with a larger chance of false eliminations. The opposite also holds true for a large confidence threshold value. In our implementation, we choose a confidence threshold of 1.15, which can have a good reduction ratio (72.1%) and guarantee a good probability of valid elimination (93.5%). Figure 18(b) shows the performance of combination reducer with different allowable ranges of power variance. We can see that the optimal range of power variance is about 1.5. Moreover, the value of α does not have a large influence to the reduction ratio due to the fact that the real scaling factor a_i is unknown.

5.6 Performance of Period Analyzer

Here we evaluate the performance of the period analyzer by comparing the True Positive Rate (TPR) and True Negative Rate (TNR) of the period analyzer in CARIN with three baselines: 1) Smokey, which uses auto-correlation to find the activity interval, and uses the variance threshold to do activity recognition [39]. 2) *cv* threshold, where we find the best *cv* threshold of the training samples, and apply it to the test data. 3) Logistic, where a logistic regression model is trained on the training set, and then the model is applied to the test data. Figure 19 shows the performance of the period analyzer with the baselines, where kNN (i.e., our period analyzer) achieves a TPR and a TNR of 97.4% and 98.3%, respectively, given a monitoring duration of 30s. Smokey has the lowest TPR and TNR because the period estimated by the auto-correlation function is not accurate. Thus, the period and variance it predicts cannot be used for further processing. Even though other baselines have become stable with a monitoring duration longer than 30s, Smokey is still oscillating around 63%. *cv* threshold and logistic can only achieve an F1 score of 76.7% and 79.1%, respectively, indicating that a linear classifier cannot be used in our scenario. The main drop for the linear model is caused by the false positives on the random head turns, where only 70% of them are recognized correctly, due to the period difference in the testing data. Moreover, when the single activities are recognized wrong, the performance shall be degraded even more. kNN achieves the best performance of all the schemes, mainly because it is more robust than the linear model, and has the tolerance to recognition error. kNN also incurs only 12ms using the k-D tree method when finding the closest distance to the testing data [4]. Figure 19 also shows the relation between monitoring duration (the time for monitoring consecutive activities and yields the period and variance) and recognition performance in terms of F1 score,

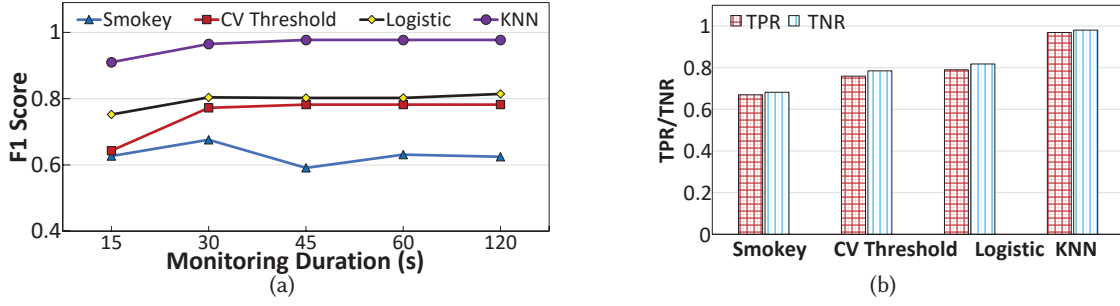


Fig. 19. Performance of period analyzer and the baselines. (a) F1 scores for different monitoring durations. (b) TPR/TNR result for fixed monitoring duration of 30s.

where a longer monitoring time yields better performance. When the monitoring duration is longer than 30s, the F1 score becomes stable because 30s is sufficient for measuring period and variance for the given CSI trace.

5.7 Robustness for Different Users and Cars

Here we first test the robustness of CARIN with 10 new users (8 males and 2 females) whose activities are not included in the offline data collection. Each user performs activities 40 times with a front passenger in the vehicle. We first test CARIN with different users in the same environment, and then test CARIN in a different car other than the one mentioned in the experiment setup.

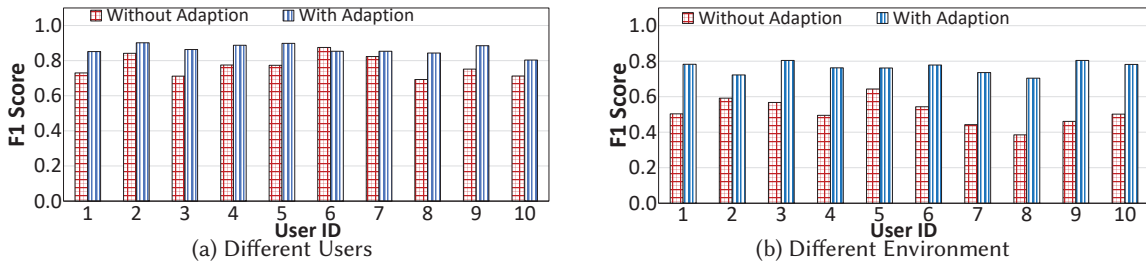


Fig. 20. Robustness test with CARIN with users and vehicles not included in the training phase. The driver activity recognition performance (F1 score) has been considerable improved with the simple adaptation algorithm.

Figure 20 shows the average F1 scores for these users in the same vehicle used for data collection (Figure 20(a)) and a different vehicle (Figure 20(b)), respectively. We can see that the overall F1 score with these users is still above 70%, even without adaptation. This result indicates that in the same vehicle, the recognition performance of the original CARIN (i.e., its F1 score) does not drop too much with new users. However, with a different vehicle, the average F1 score of CARIN drops to 53.4%. This large drop is mainly caused by both the different environment and untrained human objects. Moreover, the HMM models are trained in the same vehicle so that they may be over-fitted to one specific car as well. To address this problem and improve the robustness of CARIN, we have implemented a simple adaptation algorithm proposed in WiDrive [3], which is originally proposed to adapt the HMM activity model of the single driver scenario. Figure 20(b) shows that the average F1 score with the adaptation algorithm increases to 76.3%, which is 22.9% higher than that without the adaptation algorithm in a different car, given 40 new samples for the adaptation process. Moreover, if more test cars are available, many other machine learning methods can also be adopted to resolve the over-fitting problem, such as bagging and boosting [6]. Specifically, these methods use the data collected from a large variety of vehicles, and train multiple HMM classifiers with different preferences to different vehicle types. For online recognition, each HMM classifier yields a recognition result with its weight. Afterwards, another decision maker decides the recognition result based on the weight and result from these HMM classifiers. In this way, the variance for this “mega classifier” (including multiple HMM classifiers) can be decreased, and the robustness can be further improved.

5.8 More Interfering Activities from the Front Passenger

In this part, we extend the activity set of the front passenger, and test whether CARIN can achieve a good recognition performance. To further expand the activity set of the front passenger, we collect the CSI traces for 1) fetching the gearbox, 2) drinking/eating, and 3) adjusting window performed by the passenger. We then construct the corresponding activity combination models for the HMM classifier, and the coarse-grained features for the combination reducer. The position setup is the same as shown in Figure 10(a).

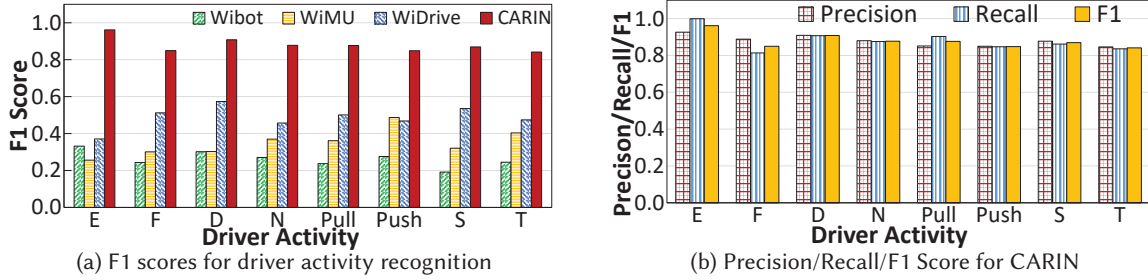


Fig. 21. Driver activity recognition performance of CARIN and the baselines with more interfering activities (e.g., fetching gearbox, drinking/eating, and adjusting window) by the front passenger.

Figure 21(a) shows the F1 scores of CARIN and the baselines. Even with more interfering activity types, CARIN can still achieve an overall F1 score of 87.9%, which is only 3.0% lower than that of the original activity set. Moreover, CARIN also performs better than the baselines with more interfering activities, especially WiMU. This is because WiMU's classifier cannot find the distinctive features for the new interfering activities, as it relies only relying on Jaccard Index for the in-car activities. Thus, WiMU can be interfered easily with activities that involve similar Doppler frequencies with different amplitudes. WiDrive can tolerate relatively more interfering activity, with a 6.4% drop of the F1 score. This is because WiDrive trains its HMM classifier for each driver activity with all traces of the same driver activity, regardless of interfering activities. Figure 21(b) shows the precision/recall and F1 scores for CARIN and the baselines, and all the precision/recall values of CARIN stay above 80% for all driver activities, while other baselines have an average decrease of 14.2%.

5.9 Real Driving Test

In this part, we test CARIN in a real driving test. Due to the fact that both the driver and the front passenger need to do the activity and it could cause danger on the road, we conduct the experiment in the same parking lot as Section 5.1.2 with a moving speed of 10mph. The driving test lasts about one and a half hour and 50 traces are collected per driver activity, with different front passenger activities (e.g., push/pull/swipe).

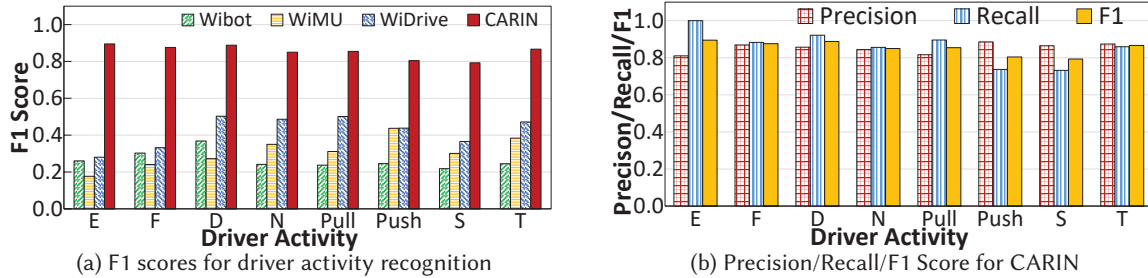


Fig. 22. Driver activity recognition performance of CARIN and the baselines in the real driving test. CARIN has a drop of 4.8% in terms of the F1 score, while the baselines drop, on average, by 17.4%.

Figure 22(a) shows the result of CARIN and baselines for the driving test. Compared with the result shown in Figure 14(a), the overall F1 score for CARIN drops to 86.1%. The drop mainly comes from the empty activity

(96.8% to 87.5%), where the driver does not do anything and the movement of surrounding can have a large impact on the CSI signal. Other than empty, CARIN can still yield a similar recognition performance to the result shown in Section 5.1.3 with only a 4.2% drop of F1 score. For the baselines, Wibot shows a larger drop of F1 score because it uses pure CSI phase information to recognize different activities, but the phase information is easy to be disturbed by moving objects other than the users (e.g., other vehicles on the road). WiMU cannot be directly applied in the car either, mainly because its classifier is simple and cannot well handle the noise caused by the moving environment. In addition, Figure 22(b) shows that CARIN can achieve an overall precision and recall of 84.6% and 86.9%, respectively, which are only 4.6% and 5.7% lower than those tested in the stationary cases.

5.10 System Overhead

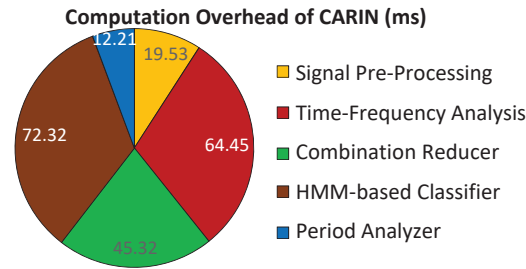


Fig. 23. System Overhead of CARIN.

Here we test the time overhead of CARIN to make sure that it can recognize activity in real-time. Figure 23 shows that the overhead of CARIN is modest: The total time overhead of CARIN is about 213ms. The combination reducer incurs 45ms but reduces more than 70% of the computation overhead in the HMM-based classifier step. The most time-consuming step is the HMM-based classifier within 73ms for the sampling rate of 1000Hz. If CARIN is implemented in DSP or other embedded systems, the overhead can be further reduced.

6 CONCLUSION

In this paper, we have proposed CARIN, a CSI-based driver activity recognition system under the interference of passengers. Unlike the state-of-art solutions for in-car activity recognition that can be used when only there is the driver in the vehicle, CARIN can recognize the driver activity even when there are one or more passengers in the vehicle performing activities together with the driver. CARIN features a combination-based solution, but overcomes the limitation of combination explosion with a novel combination reduction algorithm, which is formulated as a distance minimization problem for fast runtime recognition in the in-car scenario. CARIN also includes a period analyzer to determine the cause of an activity with a sequence of continuous movements. CARIN is evaluated in a real car with over 3000 real world in-car activity traces: The result shows that the average F1 score of CARIN is 90.9%, and can reduce the computation overhead by 72.1%.

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